

Commute Time Analysis: Interpretation, Insights, Conclusion & Recommendations

Introduction

This dashboard analyzes commute times along the Awka–Onitsha corridor, focusing on traffic, travel periods, weather, day of the week, and route direction.

KPI Overview

- Total Trips: 200
- Overall Average Commute Time: 85.15 minutes
- Average Traffic Volume: 1,660 vehicles
- Peak Average Commute Time: 116.29 minutes
- Off-Peak Average Commute Time: 69.81 minutes

Key Insight: Peak travel increases journey time by about 46 minutes, highlighting the strong effect of congestion on travel efficiency.

Business Question 1: Which Day Has the Highest Average Commute Time?

Saturday recorded the highest average commute time (93.04 minutes), while Sunday recorded the lowest (81.77 minutes).

Insight: Higher Saturday travel times may be linked to weekend commercial activities, shopping, and social events.

Recommendation: Commuters should travel earlier on Saturdays where possible, while traffic authorities should strengthen weekend traffic monitoring.

Business Question 2: How Does Traffic Level Affect Commute Time?

- High Traffic: 116.29 minutes
- Medium Traffic: 79.76 minutes
- Low Traffic: 57.17 minutes

Insight: Commute time increases significantly as traffic levels rise, making congestion the strongest factor affecting journey duration.

Recommendation: Improve congestion control, traffic management, and real-time traffic information during high-traffic periods.

Business Question 3: What Is the Difference Between Peak and Off-Peak Travel?

Peak journeys averaged 116.29 minutes, compared with 69.81 minutes during off-peak periods.

Insight: Peak travel adds approximately 46 minutes to journey time due to increased traffic demand.

Recommendation: Encourage flexible or staggered work schedules and off-peak travel where possible.

Business Question 4: Does Weather Affect Commute Time?

- Stormy: 94.14 minutes
- Rainy: 92.16 minutes
- Cloudy: 79.27 minutes
- Sunny: 74.83 minutes

Insight: Adverse weather is associated with longer journeys, likely due to reduced speeds, cautious driving, and road disruptions.

Recommendation: Allow extra travel time during storms and rainfall and improve drainage and road infrastructure.

Business Question 5: Which Route Has the Highest Average Commute Time?

- Awka → Onitsha: 86.85 minutes
- Onitsha → Awka: 81.28 minutes

Insight: Awka–Onitsha journeys take slightly longer, possibly due to differences in traffic flow and travel demand.

Recommendation: Conduct route-level analysis to identify congestion hotspots and develop targeted interventions.

Overall Conclusion

Commute time along the Awka–Onitsha corridor is mainly influenced by traffic levels and peak travel periods, with weather and day of travel also contributing. High traffic and peak periods produce the longest journeys, while low traffic and favorable weather result in shorter travel times.

Overall Recommendations

- Encourage off-peak travel where possible.
- Strengthen congestion management during peak periods.
- Improve monitoring of major congestion points.
- Prepare for increased weekend and adverse-weather traffic.
- Continuously analyze traffic patterns to support better planning and mobility.